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✓ CL-2 (BI) Assignment 3: Protein Structure Analysis

```
import numpy as np

# -----
# 1. Function to Validate Protein Sequence
# -----

def validate_sequence(sequence):

    valid_amino_acids = set("ACDEFGHIKLMNPQRSTVWY")

    for amino_acid in sequence:

        if amino_acid not in valid_amino_acids:
            return False

    return True

# -----
# 2. Function to Calculate Sequence Identity
# -----

def sequence_identity(target, template):

    length = min(len(target), len(template))

    matches = 0

    for i in range(length):

        if target[i] == template[i]:
            matches += 1

    identity = (matches / length) * 100

    return identity

# -----
# 3. Function for Simple Sequence Alignment
# -----

def simple_alignment(target, template):

    length = min(len(target), len(template))

    target_part = target[:length]
    template_part = template[:length]

    match_line = ""

    for i in range(length):

        if target_part[i] == template_part[i]:
            match_line += "|"

        else:
            match_line += "."

    return target_part, match_line, template_part
```

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# -----
# 4. Function to Predict Secondary Structure
# -----

def predict_secondary_structure(sequence):

    # Simplified classification for educational purposes

    helix_forming = set("ALMEKQRH")
    sheet_forming = set("VIFYWT")
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    structure = ""

    for amino_acid in sequence:

        if amino_acid in helix_forming:

            structure += "H"

        elif amino_acid in sheet_forming:

            structure += "E"

        else:

            structure += "C"

    return structure

# -----
# 5. Main Program
# -----

print("=" * 65)
print("          PROTEIN STRUCTURE PREDICTION")
print("          HOMOMOLOGY MODELING")
print("=" * 65)

# -----
# Take Protein Sequence as Input
# -----

sequence = input(
    "\nEnter the protein sequence: "
).strip().upper().replace(" ", "")

# -----
# Validate Input
# -----

if len(sequence) == 0:

    print("\nError: Protein sequence cannot be empty.")

    exit()

if not validate_sequence(sequence):

    print("\nError: Invalid protein sequence.")

    print(
        "Only the 20 standard amino acids "
        "are allowed."
    )

    exit()

print("\nProtein sequence is valid.")
print("Protein length:", len(sequence), "amino acids")

#MKTLLVAGAAAGTASAAPVLAADKPEMVRRLAEAGVEVVVEPGDNVVMARDPQT

```

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=====
          PROTEIN STRUCTURE PREDICTION
          HOMOMOLOGY MODELING
=====

Enter the protein sequence: MKTLLVAGAAAGTASAAPVLAADKPEMVRRLAEAGVEVVVEPGDNVVMARDPQT

Protein sequence is valid.
Protein length: 54 amino acids

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# -----
# Template Sequence
# -----

# In real homology modeling, this template would be
# obtained from a structural database such as PDB.

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template_sequence = (
    "MKTIIALSYIFCLVFADYK"
    "DDDDKGQRLIFAGKQLEDGRTLSDY"
    "VAVIDALNDS"
)

template_name = "Template_Protein_1"

print("\n" + "=" * 65)
print("                TEMPLATE INFORMATION")
print("=" * 65)

print("Template:", template_name)

print(
    "Template length:",
    len(template_sequence),
    "amino acids"
)

# -----
# Calculate Sequence Identity
# -----

identity = sequence_identity(
    sequence,
    template_sequence
)

print(
    "\nSequence Identity: {:.2f}%".format(identity)
)

# -----
# Perform Simple Alignment
# -----

target_part, match_line, template_part = simple_alignment(
    sequence,
    template_sequence
)

print("\n" + "=" * 65)
print("                SEQUENCE ALIGNMENT")
print("=" * 65)

print("\nTarget   :", target_part)

print("           ", match_line)

print("Template :", template_part)

```

```

=====
                        TEMPLATE INFORMATION
=====
Template: Template_Protein_1
Template length: 55 amino acids

Sequence Identity: 12.96%

=====
                        SEQUENCE ALIGNMENT
=====

Target   : MKTLLVAGAAAGTASAAPVLAADKPEMVRRLAEAGVEVVVEPGDNVVMRDPQT
          |||.....|.....||.....|.....|.....
Template : MKTIIALSYIFCLVFADYKDDDDKGQRLIFAGKQLEDGRTLSDYVAVIDALNDS

```

```

# -----
# Predict Secondary Structure
# -----

secondary_structure = predict_secondary_structure(
    sequence
)

print("\n" + "=" * 65)

```

```

print("                PREDICTED SECONDARY STRUCTURE")
print("=" * 65)

print("\nH = Alpha Helix")
print("E = Beta Sheet")
print("C = Coil")

print("\nProtein Sequence:")
print(sequence)

print("\nPredicted Structure:")
print(secondary_structure)

# -----
# Estimate Modeling Confidence
# -----

if identity >= 50:

    confidence = "High"

elif identity >= 30:

    confidence = "Moderate"

else:

    confidence = "Low"

print("\nEstimated Modeling Confidence:", confidence)

```

```

=====
                PREDICTED SECONDARY STRUCTURE
=====

H = Alpha Helix
E = Beta Sheet
C = Coil

Protein Sequence:
MKTLLVAGAAAGTASAAPVLAADKPEMVRRLAEAGVEVVVEPGDNVVMARDPQT

Predicted Structure:
HHEHHEHCHHHCEHCHHCEHHHCHCHHEHHHHHCEHEEEHCCCEEHHHCHE

Estimated Modeling Confidence: Low

```

```

# -----
# Structural Representation
# -----

print("\n" + "=" * 65)
print("                STRUCTURAL REPRESENTATION")
print("=" * 65)

print("\nPosition  Amino Acid  Structure")

print("-" * 35)

for i, amino_acid in enumerate(sequence):

    structure = secondary_structure[i]

    if structure == "H":

        symbol = "Alpha Helix"

    elif structure == "E":

        symbol = "Beta Sheet"

    else:

        symbol = "Coil"

```

```

print(
    "{:8d}  {:11s}  {}".format(
        i + 1,
        amino_acid,
        symbol
    )
)

```

```

=====
Position  Amino Acid  Structure
-----
    1  M          Alpha Helix
    2  K          Alpha Helix
    3  T          Beta Sheet
    4  L          Alpha Helix
    5  L          Alpha Helix
    6  V          Beta Sheet
    7  A          Alpha Helix
    8  G          Coil
    9  A          Alpha Helix
   10  A          Alpha Helix
   11  A          Alpha Helix
   12  G          Coil
   13  T          Beta Sheet
   14  A          Alpha Helix
   15  S          Coil
   16  A          Alpha Helix
   17  A          Alpha Helix
   18  P          Coil
   19  V          Beta Sheet
   20  L          Alpha Helix
   21  A          Alpha Helix
   22  A          Alpha Helix
   23  D          Coil
   24  K          Alpha Helix
   25  P          Coil
   26  E          Alpha Helix
   27  M          Alpha Helix
   28  V          Beta Sheet
   29  R          Alpha Helix
   30  R          Alpha Helix
   31  L          Alpha Helix
   32  A          Alpha Helix
   33  E          Alpha Helix
   34  A          Alpha Helix
   35  G          Coil
   36  V          Beta Sheet
   37  E          Alpha Helix
   38  V          Beta Sheet
   39  V          Beta Sheet
   40  V          Beta Sheet
   41  E          Alpha Helix
   42  P          Coil
   43  G          Coil
   44  D          Coil
   45  N          Coil
   46  V          Beta Sheet
   47  V          Beta Sheet
   48  A          Alpha Helix
   49  M          Alpha Helix
   50  R          Alpha Helix
   51  D          Coil
   52  P          Coil
   53  Q          Alpha Helix
   54  T          Beta Sheet

```

```

# -----
# Final Result
# -----

print("\n" + "=" * 65)
print("                FINAL RESULT")
print("=" * 65)

print(
    "\nTarget Protein Length:",
    len(sequence),
    "amino acids"
)

print("Template:", template_name)

print(
    "Sequence Identity: {:.2f}%".format(identity)
)

```

```
print("Modeling Confidence:", confidence)

print("\nProtein structure prediction completed.")

print("=" * 65)
```

```
=====
                        FINAL RESULT
=====

Target Protein Length: 54 amino acids
Template: Template_Protein_1
Sequence Identity: 12.96%
Modeling Confidence: Low

Protein structure prediction completed.
=====
```